

Data Types

Data types represents the type of data.

Built-In Data Types

- int
- float
- str
- bool
- complex
- list
- tuple
- set
- frozenset
- dict

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type() Function in Python

The type() function is a built-in function in Python used to find out the data type of a variable or value.

It helps you understand what kind of data you're working with — like int, float, str, list, etc.

```
a = True
print(a)
print(type(a))
print(type(True))
```

int data type

```
1 a=10
2 print(a)
3 print(type(a))
```

float data type

```
1 a=10.4
2 print(a)
3 print(type(a))
```

str data type

```
1 name="Rajat"  
2 print(name)  
3 print(type(name))  
4
```

bool data type

```
1 x=True  
2 print(x)  
3 print(type(x))  
4
```

complex data type

```
1 x = 10 + 5j
2 print(x)
3 print(type(x))
```

```
1 x = 10 + 5j
2 print(x.real)
3 print(x.imag)
```

Type Casting (Type Conversion) in Python

Type casting (or type conversion) means changing the data type of a value or variable from one type to another.

Python provides several built-in functions to perform this conversion easily.

- int
- float
- str
- bool
- complex

int() function

To convert another data type into an integer except complex numbers.

```
1  a=10.3
2  print(a)
3  print(type(a))
4  b=int(a)
5  print(b)
6  print(type(b))
```

float() function

To convert another data type into floating-point number except complex numbers.

```
1  a=True
2  print(a)
3  print(type(a))
4  b=float(a)
5  print(b)
6  print(type(b))
7
```


str() function

To convert another data type into a string. The str() function can also convert a complex number into a string.

```
1  a=10
2  print(a)
3  print(type(a))
4  b=str(a)
5  print(b)
6  print(type(b))
7
```

bool() function

To convert another data type into a bool. The bool() function can also convert a complex number into a Boolean.

```
1  a=1
2  print(a)
3  print(type(a))
4  b=bool(a)
5  print(b)
6  print(type(b))
```

complex() function

To convert another data type into a complex.

```
1  a=10
2  b=20
3  c=complex(a,b)
4  print(c)
5  print(type(c))
6
```

Other Conversion Functions

- `bin` → To convert the number into binary.
- `oct` → To convert the number into octal.
- `hex` → To convert the number into hexadecimal.

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